



8051 - MIRI Spectroscopic survey at $z \sim 10$: Insights into the Nature of Primordial Galaxies

Cycle: 4, Proposal Category: GO

INVESTIGATORS

<i>Name</i>	<i>Institution</i>
Dr. Javier Alvarez-Marquez (PI) (ESA Member)	Centro de Astrobiologia - CAB
Dr. Sarah Kendrew (CoI) (ESA Member) (US Admin CoI)	Space Telescope Science Institute - ESA - JWST
Luis Colina Robledo (CoI) (ESA Member) (CoPI)	Centro de Astrobiologia - CAB
Prof. Goeran Oestlin (CoI) (ESA Member)	Stockholm University
Dr. Arjan Bik (CoI) (ESA Member)	Stockholm University
Dr. Tiger Hsiao (CoI)	University of Texas at Austin
Dr. Abdurro'uf Abdurro'uf (CoI)	Indiana University System
Dr. Seiji Fujimoto (CoI) (CSA Member)	University of Toronto
Dr. Dan Coe (CoI) (ESA Member)	Space Telescope Science Institute - ESA - JWST
Dr. Leindert Boogaard (CoI) (ESA Member)	Universiteit Leiden
Dr. Pierluigi Rinaldi (CoI)	Space Telescope Science Institute
Dr. Takuya Hashimoto (CoI)	University of Tsukuba
Dr. Jorge Zavala (CoI)	University of Massachusetts - Amherst
Dr. Yoshinobu Fudamoto (CoI)	Chiba University
Rui Marques-Chaves (CoI) (ESA Member)	University of Geneva, Department of Astronomy
Prof. Pablo G. Perez Gonzalez (CoI) (ESA Member)	Centro de Astrobiologia - CAB
Dr. Jens Melinder (CoI) (ESA Member)	Stockholm University
Carlota Prieto (CoI) (ESA Member)	Centro de Astrobiologia - CAB
Msr. Carmen Blanco (CoI) (ESA Member)	Centro de Astrobiologia - CAB
Dr. Alejandro Crespo Gomez (CoI)	Space Telescope Science Institute
Prof. Akio Inoue (CoI)	Waseda University

JWST Proposal 8051 (Created: Wednesday, May 20, 2026, 3:00:11PM Eastern Standard Time) - Overview

Name	Institution
Dr. Bruno Rodriguez Del Pino (CoI) (ESA Member)	Centro de Astrobiologia - CAB
Macarena Garcia Marin (CoI) (ESA Member)	Space Telescope Science Institute - ESA - JWST
Gillian Wright (CoI) (ESA Member)	United Kingdom Astronomy Technology Centre
Dr. Yuma Sugahara (CoI)	Waseda University
Dr. Yuichi Harikane (CoI)	University of Tokyo, Institute of Cosmic Ray Research
Dr. Paola Santini (CoI) (ESA Member)	INAF - Osservatorio Astronomico di Roma
Dr. Marco Castellano (CoI) (ESA Member)	INAF - Osservatorio Astronomico di Roma
Mr. Lorenzo Napolitano (CoI) (ESA Member)	INAF - Osservatorio Astronomico di Roma
Dr. Adriano Fontana (CoI) (ESA Member)	INAF - Osservatorio Astronomico di Roma
Prof. Tommaso L. Treu (CoI)	University of California - Los Angeles
Prof. Karina Caputi (CoI) (ESA Member)	Kapteyn Astronomical Institute
Mr. Danial Langeroodi (CoI) (ESA Member)	University of Copenhagen, Niels Bohr Institute
Dr. Edoardo Iani (CoI) (ESA Member)	Institute of Science and Technology Austria

OBSERVATIONS

Folder	Observation	Label	Observing Template	Science Target
GHZ7				
	1		MIRI Low Resolution Spectroscopy	(1) GHZ7
	2		MIRI Low Resolution Spectroscopy	(1) GHZ7
GHZ9-2				
	22		MIRI Low Resolution Spectroscopy	(17) GHZ9-2
Uncover-37126				
	5		MIRI Low Resolution Spectroscopy	(2) Uncover-37126
	6		MIRI Low Resolution Spectroscopy	(2) Uncover-37126
GHZ8				
	7		MIRI Low Resolution Spectroscopy	(3) GHZ8
	8		MIRI Low Resolution Spectroscopy	(3) GHZ8
GHZ9				
	9		MIRI Low Resolution Spectroscopy	(4) GHZ9
	10		MIRI Low Resolution Spectroscopy	(4) GHZ9
Uncover-26185				
	11		MIRI Low Resolution Spectroscopy	(5) Uncover-26185

Folder	Observation	Label	Observing Template	Science Target
	12		MIRI Low Resolution Spectroscopy	(5) Uncover-26185
	13		MIRI Low Resolution Spectroscopy	(5) Uncover-26185
CEERS_99715				
	14		MIRI Low Resolution Spectroscopy	(6) CEERS_99715
Uncover-13151				
	15		MIRI Low Resolution Spectroscopy	(7) Uncover-13151
	16		MIRI Low Resolution Spectroscopy	(7) Uncover-13151
GHZ1				
	17		MIRI Low Resolution Spectroscopy	(8) GHZ1
GN-55757-1055757				
	18		MIRI Low Resolution Spectroscopy	(9) GN-55757-1055757
	19		MIRI Low Resolution Spectroscopy	(9) GN-55757-1055757
GS-20088041-207868				
	20		MIRI Low Resolution Spectroscopy	(10) GS-20088041-207868
	21		MIRI Low Resolution Spectroscopy	(10) GS-20088041-207868

ABSTRACT

One of the most unexpected discoveries of JWST has been the large number of luminous galaxies at very early epochs in the Universe ($z > 10$), far exceeding the predictions for current galaxy formation models. While several astrophysical scenarios have been proposed, the nature and formation of these galaxies remain largely unknown and a subject of active debate. The primary limitation comes from the narrow rest-frame spectral coverage ($< 0.4\mu\text{m}$) of the NIRCам and NIRSpec instruments.

This survey aims to address these gaps by leveraging MIRI LRS spectroscopy to detect key optical emission lines, such as H β + [OIII]4960,5008 and H α , and the underline optical continuum. These observations will constrain critical astrophysical parameters, including SFR, burstiness, ionizing production efficiency, stellar mass, metallicity, and also, establish the dominant ionizing source (starburst and/or AGN) in these primordial galaxies.

The survey targets a sample of ten galaxies at redshift of ~ 10 , including two X-ray AGN candidates. The sample spans a wide range in UV absolute magnitudes (-17.6 to -20.5), extending well below the characteristic M^*_{UV} (-20.6 mag) at $z \sim 10$. This proposal will provide critical insights into the nature of primordial galaxies, offering a comprehensive view of the build up of stars, metals, and dust in the early Universe, and the processes driving

the formation of luminous galaxies in the pre-Reionization Epoch.

OBSERVING DESCRIPTION

This survey requests deep MIRI LRS spectroscopy for a sample of ten galaxies at $z \sim 10$. MIRI LRS will cover the rest-frame optical spectrum, ensuring the detection of strong optical emission lines, such as H β + $[\text{OIII}]$ 4960,5008 and H α , and the underline optical continuum.

On-source exposure times ranging from 23ks to 60ks are requested depending on the expected fluxes of H β + $[\text{OIII}]$ and H α emission lines. A setup with 149 groups using the FASTR1 readout mode, the default for LRS, is chosen. The selection of 149 groups ensures that each integration is below 500 seconds, mitigating the effects of cosmic ray showers. The number of integrations and observations for each galaxy in the survey is adjusted accordingly to match the required exposure times.

A 2-column mosaic strategy is used to effectively double the number of dither pointings. The LRS template only offers a 2-point dither strategy; we chose our mosaic and column-overlap strategy to ensure the targets are placed at 4 different slit positions rather than just 2, providing better mitigation for bad or hot pixels and an improved reconstruction of the background for background subtraction. This strategy was informed by past experience with LRS data.

Target acquisition (TA) will be used where a suitable GAIA DR3 star is available within the permitted separation. For galaxies without a nearby GAIA star, a blind pointing approach will be adopted, as the JWST's pointing accuracy ($\sim 0.1''$) is well within the LRS slit dimension ($4.7'' \times 0.51''$).

To maximize sensitivity and efficiency, this proposal requires a background-limited setting. Then, we have selected a "background no more than 20% above the minimum".

V3PA constraints have been requested in some galaxies of the sample to (i) avoid bright sources within the MIRI LRS slit field of view, and (ii) align the slit along the major axes of extended and elongated sources. However, most of the galaxies are expected to be point-like for the LRS PSF.

Based on calculations from the JWST ETC, MIRI LRS will detect an integrated H α line at a minimum significance level of 5-7 σ across all galaxies in the sample. The H β + $[\text{OIII}]$ lines will be detected with higher significance, i.e. integrated SNR between 7 (2% Z_{sun}) and 30 (15% Z_{sun}). ETC also predicts a 4-7 σ detection of the rest-frame optical continuum at ~ 4900 angstroms (only for $z > 10$) and ~ 5900 angstroms (all redshift) stacking several spectral elements.

The requested exclusive access period is 6 months to maximize and accelerate the scientific outcomes of this program.

Proposal 8051 - Targets - MIRI Spectroscopic survey at $z \sim 10$: Insights into the Nature of Primordial Galaxies

(8)	GHZ1	RA: 00 14 2.8630 (3.5119292d) Dec: -30 22 18.69 (-30.37186d) Equinox: J2000	
Comments: Category=Galaxy Description=[Compact galaxies, Emission line galaxies, High-redshift galaxies, Starburst galaxies] Extended=NO			
(9)	GN-55757-1055757	RA: 12 36 52.2449 (189.2176871d) Dec: +62 11 58.17 (62.19949d) Equinox: J2000	
Comments: Category=Galaxy Description=[Compact galaxies, Emission line galaxies, High-redshift galaxies, Starburst galaxies] Extended=NO			
(10)	GS-20088041-207868	RA: 03 32 42.1242 (53.1755175d) Dec: -27 46 50.29 (-27.78064d) Equinox: J2000	
Comments: Category=Galaxy Description=[Compact galaxies, Emission line galaxies, High-redshift galaxies, Starburst galaxies] Extended=NO			
(11)	Uncover-37126-TA	RA: 00 14 19.7527 (3.5823029d) Dec: -30 21 16.29 (-30.35453d) Equinox: J2000	Proper Motion RA: -2.1604 mas/yr Proper Motion Dec: -4.6065 mas/yr Parallax: 0.0016933163940464273" Epoch of Position: 2016
Comments: Category=Star Description=[A stars] Extended=NO			
(12)	Uncover-26185-TA	RA: 00 14 16.3855 (3.5682729d) Dec: -30 22 2.73 (-30.36743d) Equinox: J2000	Proper Motion RA: 37.1398 mas/yr Proper Motion Dec: -2.6180 mas/yr Parallax: 0.0036153262837062314" Epoch of Position: 2016
Comments: Category=Star Description=[A stars] Extended=NO			
(13)	Uncover-13151-TA	RA: 00 14 21.7948 (3.5908117d) Dec: -30 24 9.02 (-30.40251d) Equinox: J2000	Proper Motion RA: -5.8384 mas/yr Proper Motion Dec: -7.6242 mas/yr Epoch of Position: 2016
Comments: Category=Star Description=[A stars] Extended=NO			
(14)	GHZ78-TA	RA: 00 13 47.0682 (3.4461175d) Dec: -30 19 36.49 (-30.32680d) Equinox: J2000	Proper Motion RA: -8.2626 mas/yr Proper Motion Dec: -2.6433 mas/yr Parallax: 0.0014463" Epoch of Position: 2016
Comments: Category=Star Description=[A stars] Extended=NO			

Proposal 8051 - Targets - MIRI Spectroscopic survey at $z \sim 10$: Insights into the Nature of Primordial Galaxies

(15)	GHZ1-TA	RA: 00 14 5.0291 (3.5209546d) Dec: -30 22 22.22 (-30.37284d) Equinox: J2000	Proper Motion RA: 5.2250 mas/yr Proper Motion Dec: -6.4074 mas/yr Parallax: 0.0004943" Epoch of Position: 2016
Comments: Category=Star Description=[A stars] Extended=NO			
(16)	GHZ9-TA	RA: 00 13 52.6421 (3.4693421d) Dec: -30 20 16.27 (-30.33785d) Equinox: J2000	Proper Motion RA: -7.2989 mas/yr Proper Motion Dec: -8.7625 mas/yr Parallax: 0.0029187" Epoch of Position: 2016
Comments: Category=Star Description=[A stars] Extended=NO			
(17)	GHZ9-2	RA: 00 13 54.9026 (3.4787608d) Dec: -30 20 43.86 (-30.34552d) Equinox: J2000	
Comments: Category=Galaxy Description=[Compact galaxies, Emission line galaxies, High-redshift galaxies, Starburst galaxies] Extended=NO			

Proposal 8051 - Observation 1 - MIRI Spectroscopic survey at z~10: Insights into the Nature of Primordial Galaxies

Observation	Proposal 8051, Observation 1 Wed May 20 20:00:11 GMT 2026									
	Diagnostic Status: Warning Observing Template: MIRI Low Resolution Spectroscopy									
Diagnostics	(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.									
Fixed Targets	#	Name	Target Coordinates		Targ. Coord. Corrections			Miscellaneous		
	(1)	GHZ7	RA: 00 13 48.3286 (3.4513692d) Dec: -30 19 14.58 (-30.32072d) Equinox: J2000							
	<i>Comments:</i> <i>Category=Galaxy</i> <i>Description=[Compact galaxies, Emission line galaxies, High-redshift galaxies, Starburst galaxies]</i> <i>Extended=NO</i>									
Acquisition	#									Target
	1									NONE
Template	AcqFilter	Subarray				Obtain Verification Image?				
	F560W	FULL				true				
Mosaic	Rows	Columns	Row Overlap %	Column Overlap %	Row shift (deg)	Column shift (deg)	Tile Order			
	1	2	0.0	80.0	0.0	0.0	DEFAULT			
Dithers	#	Dither Type	No. Spectral Steps	Spectral Step Offset	No. Spatial Steps	Spatial Step Offset				
	1	ALONG SLIT NOD								
Pointing Verification	#	PV Readout Pattern	PV Groups/Int	PV Integrations/Exp	PV Total Integrations	PV Exposures/Dith	PV Total Dithers	PV Total Exposure Time	Optional ETC ID	Filter
	1	FASTR1	34	3	3	1	1	288.604		F560W

Proposal 8051 - Observation 1 - MIRI Spectroscopic survey at $z \sim 10$: Insights into the Nature of Primordial Galaxies

Spectral Elements	#	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Exposures/Dith	Total Dithers	Total Exposure Time	Optional ETC ID
	1	FASTR1	149	11	22	1	2	9152.082	220843
Special Requirements	Background Limited. Background no more than 20th percentile above minimum								
	Group Observations 1, 2, Non-interruptible								

Proposal 8051 - Observation 2 - MIRI Spectroscopic survey at z~10: Insights into the Nature of Primordial Galaxies

Observation	Proposal 8051, Observation 2 Wed May 20 20:00:11 GMT 2026									
	Diagnostic Status: Warning Observing Template: MIRI Low Resolution Spectroscopy									
Diagnostics	(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.									
Fixed Targets	#	Name	Target Coordinates		Targ. Coord. Corrections			Miscellaneous		
	(1)	GHZ7	RA: 00 13 48.3286 (3.4513692d) Dec: -30 19 14.58 (-30.32072d) Equinox: J2000							
	<i>Comments:</i> <i>Category=Galaxy</i> <i>Description=[Compact galaxies, Emission line galaxies, High-redshift galaxies, Starburst galaxies]</i> <i>Extended=NO</i>									
Acquisition	#									Target
	1									NONE
Template	AcqFilter	Subarray				Obtain Verification Image?				
	F560W	FULL				true				
Mosaic	Rows	Columns	Row Overlap %	Column Overlap %	Row shift (deg)	Column shift (deg)	Tile Order			
	1	2	0.0	80.0	0.0	0.0	DEFAULT			
Dithers	#	Dither Type	No. Spectral Steps	Spectral Step Offset	No. Spatial Steps	Spatial Step Offset				
	1	ALONG SLIT NOD								
Pointing Verification	#	PV Readout Pattern	PV Groups/Int	PV Integrations/Exp	PV Total Integrations	PV Exposures/Dith	PV Total Dithers	PV Total Exposure Time	Optional ETC ID	Filter
	1	FASTR1	34	3	3	1	1	288.604		F560W

Proposal 8051 - Observation 2 - MIRI Spectroscopic survey at z~10: Insights into the Nature of Primordial Galaxies

Spectral Elements	#	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Exposures/Dith	Total Dithers	Total Exposure Time	Optional ETC ID
	1	FASTR1	149	11	22	1	2	9152.082	220843
Special Requirements	Background Limited. Background no more than 20th percentile above minimum								
	Group Observations 1, 2, Non-interruptible								

Proposal 8051 - Observation 22 - MIRI Spectroscopic survey at z~10: Insights into the Nature of Primordial Galaxies

Observation	Proposal 8051, Observation 22									Wed May 20 20:00:11 GMT 2026
	Diagnostic Status: Warning									
	Observing Template: MIRI Low Resolution Spectroscopy									
Diagnostics	(Visit 22:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.									
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections		Miscellaneous		
	(17)	GHZ9-2	RA: 00 13 54.9026 (3.4787608d) Dec: -30 20 43.86 (-30.34552d) Equinox: J2000							
	Comments: Category=Galaxy Description=[Compact galaxies, Emission line galaxies, High-redshift galaxies, Starburst galaxies] Extended=NO									
Acquisition	#	Target								
	1	NONE								
Template	AcqFilter	Subarray				Obtain Verification Image?				
	F560W	FULL				true				
Mosaic	Rows	Columns	Row Overlap %		Column Overlap %		Row shift (deg)		Column shift (deg)	Tile Order
	1	2	0.0		80.0		0.0		0.0	DEFAULT
Dithers	#	Dither Type		No. Spectral Steps		Spectral Step Offset		No. Spatial Steps		Spatial Step Offset
	1	ALONG SLIT NOD								
Pointing Verification	#	PV Readout Pattern	PV Groups/Int	PV Integrations/Exp	PV Total Integrations	PV Exposures/Dith	PV Total Dithers	PV Total Exposure Time	Optional ETC ID	Filter
	1	FASTR1	34	3	3	1	1	288.604		F560W

Proposal 8051 - Observation 22 - MIRI Spectroscopic survey at z~10: Insights into the Nature of Primordial Galaxies

Spectral Elements	#	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Exposures/Dith	Total Dithers	Total Exposure Time	Optional ETC ID
	1	FASTR1	149	15	30	1	2	12482.13	220843
Special Requirements	Background Limited. Background no more than 20th percentile above minimum								

Proposal 8051 - Observation 5 - MIRI Spectroscopic survey at z~10: Insights into the Nature of Primordial Galaxies

Observation	Proposal 8051, Observation 5									Wed May 20 20:00:11 GMT 2026
	Diagnostic Status: Warning									
	Observing Template: MIRI Low Resolution Spectroscopy									
Diagnostics	(Visit 5:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.									
Fixed Targets	#	Name	Target Coordinates		Targ. Coord. Corrections			Miscellaneous		
	(2)	Uncover-37126	RA: 00 14 21.6266 (3.5901108d) Dec: -30 21 35.07 (-30.35974d) Equinox: J2000							
	Comments: Category=Galaxy Description=[Compact galaxies, Emission line galaxies, High-redshift galaxies, Starburst galaxies] Extended=NO									
Acquisition	#	Target	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID	
	1	11 Uncover-37126-TA	F560W	FAST	4	1	1	11.1	221225	
Template	Subarray				Obtain Verification Image?					
	FULL				true					
Mosaic	Rows	Columns	Row Overlap %		Column Overlap %	Row shift (deg)	Column shift (deg)		Tile Order	
	1	2	0.0		80.0	0.0	0.0		DEFAULT	
Dithers	#	Dither Type		No. Spectral Steps		Spectral Step Offset		No. Spatial Steps		Spatial Step Offset
	1	ALONG SLIT NOD								
Pointing Verification	#	PV Readout Pattern	PV Groups/Int	PV Integrations/Exp	PV Total Integrations	PV Exposures/Dith	PV Total Dithers	PV Total Exposure Time	Optional ETC ID	Filter
	1	FASTR1	34	3	3	1	1	288.604		F560W

Proposal 8051 - Observation 5 - MIRI Spectroscopic survey at $z \sim 10$: Insights into the Nature of Primordial Galaxies

Spectral Elements	#	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Exposures/Dith	Total Dithers	Total Exposure Time	Optional ETC ID
	1	FASTR1	149	12	24	1	2	9984.594	220843
Special Requirements	Aperture PA Range 29.75797 to 74.75797 Degrees (V3 25.0 to 70.0) Aperture PA Range 234.75797 to 274.75797 Degrees (V3 230.0 to 270.0) Background Limited. Background no more than 20th percentile above minimum Group Observations 5, 6, Non-interruptible								

Proposal 8051 - Observation 6 - MIRI Spectroscopic survey at z~10: Insights into the Nature of Primordial Galaxies

Observation	Proposal 8051, Observation 6									Wed May 20 20:00:12 GMT 2026
	Diagnostic Status: Warning									
	Observing Template: MIRI Low Resolution Spectroscopy									
Diagnostics	(Visit 6:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.									
Fixed Targets	#	Name	Target Coordinates		Targ. Coord. Corrections			Miscellaneous		
	(2)	Uncover-37126	RA: 00 14 21.6266 (3.5901108d) Dec: -30 21 35.07 (-30.35974d) Equinox: J2000							
	Comments: Category=Galaxy Description=[Compact galaxies, Emission line galaxies, High-redshift galaxies, Starburst galaxies] Extended=NO									
Acquisition	#	Target	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID	
	1	11 Uncover-37126-TA	F560W	FAST	4	1	1	11.1	221225	
Template	Subarray					Obtain Verification Image?				
	FULL					true				
Mosaic	Rows	Columns	Row Overlap %		Column Overlap %	Row shift (deg)	Column shift (deg)		Tile Order	
	1	2	0.0		80.0	0.0	0.0		DEFAULT	
Dithers	#	Dither Type		No. Spectral Steps		Spectral Step Offset		No. Spatial Steps		Spatial Step Offset
	1	ALONG SLIT NOD								
Pointing Verification	#	PV Readout Pattern	PV Groups/Int	PV Integrations/Exp	PV Total Integrations	PV Exposures/Dith	PV Total Dithers	PV Total Exposure Time	Optional ETC ID	Filter
	1	FASTR1	34	3	3	1	1	288.604		F560W

Proposal 8051 - Observation 6 - MIRI Spectroscopic survey at $z \sim 10$: Insights into the Nature of Primordial Galaxies

Spectral Elements	#	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Exposures/Dith	Total Dithers	Total Exposure Time	Optional ETC ID
	1	FASTR1	149	12	24	1	2	9984.594	220843
Special Requirements	Aperture PA Range 29.75797 to 74.75797 Degrees (V3 25.0 to 70.0) Aperture PA Range 234.75797 to 274.75797 Degrees (V3 230.0 to 270.0) Background Limited. Background no more than 20th percentile above minimum Group Observations 5, 6, Non-interruptible								

Proposal 8051 - Observation 7 - MIRI Spectroscopic survey at z~10: Insights into the Nature of Primordial Galaxies

Observation	Proposal 8051, Observation 7									Wed May 20 20:00:12 GMT 2026
	Diagnostic Status: Warning									
	Observing Template: MIRI Low Resolution Spectroscopy									
Diagnostics	(Visit 7:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.									
Fixed Targets	#	Name	Target Coordinates		Targ. Coord. Corrections			Miscellaneous		
	(3)	GHZ8	RA: 00 13 48.3427 (3.4514279d) Dec: -30 19 18.47 (-30.32180d) Equinox: J2000							
	Comments: Category=Galaxy Description=[Compact galaxies, Emission line galaxies, High-redshift galaxies, Starburst galaxies] Extended=NO									
Acquisition	#	Target	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID	
	1	14 GHZ78-TA	F560W	FAST	4	1	1	11.1	221225	
Template	Subarray				Obtain Verification Image?					
	FULL				true					
Mosaic	Rows	Columns	Row Overlap %		Column Overlap %	Row shift (deg)		Column shift (deg)	Tile Order	
	1	2	0.0		80.0	0.0		0.0	DEFAULT	
Dithers	#	Dither Type		No. Spectral Steps		Spectral Step Offset		No. Spatial Steps		Spatial Step Offset
	1	ALONG SLIT NOD								
Pointing Verification	#	PV Readout Pattern	PV Groups/Int	PV Integrations/Exp	PV Total Integrations	PV Exposures/Dith	PV Total Dithers	PV Total Exposure Time	Optional ETC ID	Filter
	1	FASTR1	34	3	3	1	1	288.604		F560W

Proposal 8051 - Observation 7 - MIRI Spectroscopic survey at $z \sim 10$: Insights into the Nature of Primordial Galaxies

Spectral Elements	#	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Exposures/Dith	Total Dithers	Total Exposure Time	Optional ETC ID
	1	FASTR1	149	8	16	1	2	6654.546	220843
Special Requirements	Background Limited. Background no more than 20th percentile above minimum								
	Group Observations 7, 8, Non-interruptible								

Proposal 8051 - Observation 8 - MIRI Spectroscopic survey at z~10: Insights into the Nature of Primordial Galaxies

Observation	Proposal 8051, Observation 8									Wed May 20 20:00:12 GMT 2026
	Diagnostic Status: Warning									
	Observing Template: MIRI Low Resolution Spectroscopy									
Diagnostics	(Visit 8:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.									
Fixed Targets	#	Name	Target Coordinates		Targ. Coord. Corrections			Miscellaneous		
	(3)	GHZ8	RA: 00 13 48.3427 (3.4514279d) Dec: -30 19 18.47 (-30.32180d) Equinox: J2000							
	Comments: Category=Galaxy Description=[Compact galaxies, Emission line galaxies, High-redshift galaxies, Starburst galaxies] Extended=NO									
Acquisition	#	Target	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID	
	1	14 GHZ78-TA	F560W	FAST	4	1	1	11.1	221225	
Template	Subarray					Obtain Verification Image?				
	FULL					true				
Mosaic	Rows	Columns	Row Overlap %		Column Overlap %	Row shift (deg)		Column shift (deg)	Tile Order	
	1	2	0.0		80.0	0.0		0.0	DEFAULT	
Dithers	#	Dither Type		No. Spectral Steps		Spectral Step Offset		No. Spatial Steps		Spatial Step Offset
	1	ALONG SLIT NOD								
Pointing Verification	#	PV Readout Pattern	PV Groups/Int	PV Integrations/Exp	PV Total Integrations	PV Exposures/Dith	PV Total Dithers	PV Total Exposure Time	Optional ETC ID	Filter
	1	FASTR1	34	3	3	1	1	288.604		F560W

Proposal 8051 - Observation 8 - MIRI Spectroscopic survey at z~10: Insights into the Nature of Primordial Galaxies

Spectral Elements	#	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Exposures/Dith	Total Dithers	Total Exposure Time	Optional ETC ID
	1	FASTR1	149	8	16	1	2	6654.546	220843
Special Requirements	Background Limited. Background no more than 20th percentile above minimum								
	Group Observations 7, 8, Non-interruptible								

Proposal 8051 - Observation 9 - MIRI Spectroscopic survey at z~10: Insights into the Nature of Primordial Galaxies

Observation	Proposal 8051, Observation 9									Wed May 20 20:00:12 GMT 2026
	Diagnostic Status: Warning									
	Observing Template: MIRI Low Resolution Spectroscopy									
Diagnostics	(Visit 9:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.									
Fixed Targets	#	Name	Target Coordinates		Targ. Coord. Corrections		Miscellaneous			
	(4)	GHZ9	RA: 00 13 54.9026 (3.4787608d) Dec: -30 20 43.86 (-30.34552d) Equinox: J2000							
	Comments: Category=Galaxy Description=[Compact galaxies, Emission line galaxies, High-redshift galaxies, Starburst galaxies] Extended=NO									
Acquisition	#	Target	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID	
	1	16 GHZ9-TA	FND	FAST	10	1	1	27.75	221225	
Template	Subarray				Obtain Verification Image?					
	FULL				true					
Mosaic	Rows	Columns	Row Overlap %		Column Overlap %	Row shift (deg)	Column shift (deg)		Tile Order	
	1	2	0.0		80.0	0.0	0.0		DEFAULT	
Dithers	#	Dither Type		No. Spectral Steps		Spectral Step Offset		No. Spatial Steps		Spatial Step Offset
	1	ALONG SLIT NOD								
Pointing Verification	#	PV Readout Pattern	PV Groups/Int	PV Integrations/Exp	PV Total Integrations	PV Exposures/Dith	PV Total Dithers	PV Total Exposure Time	Optional ETC ID	Filter
	1	FASTR1	34	3	3	1	1	288.604		F560W

Proposal 8051 - Observation 9 - MIRI Spectroscopic survey at z~10: Insights into the Nature of Primordial Galaxies

Spectral Elements	#	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Exposures/Dith	Total Dithers	Total Exposure Time	Optional ETC ID
	1	FASTR1	149	8	16	1	2	6654.546	220843
Special Requirements	Background Limited. Background no more than 20th percentile above minimum								
	Group Observations 9, 10, Non-interruptible								

Proposal 8051 - Observation 10 - MIRI Spectroscopic survey at z~10: Insights into the Nature of Primordial Galaxies

Observation	Proposal 8051, Observation 10									Wed May 20 20:00:12 GMT 2026
	Diagnostic Status: Warning									
	Observing Template: MIRI Low Resolution Spectroscopy									
Diagnostics	(Visit 10:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.									
Fixed Targets	#	Name	Target Coordinates		Targ. Coord. Corrections			Miscellaneous		
	(4)	GHZ9	RA: 00 13 54.9026 (3.4787608d) Dec: -30 20 43.86 (-30.34552d) Equinox: J2000							
	Comments: Category=Galaxy Description=[Compact galaxies, Emission line galaxies, High-redshift galaxies, Starburst galaxies] Extended=NO									
Acquisition	#	Target	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID	
	1	16 GHZ9-TA	FND	FAST	10	1	1	27.75	221225	
Template	Subarray					Obtain Verification Image?				
	FULL					true				
Mosaic	Rows	Columns	Row Overlap %		Column Overlap %	Row shift (deg)	Column shift (deg)		Tile Order	
	1	2	0.0		80.0	0.0	0.0		DEFAULT	
Dithers	#	Dither Type		No. Spectral Steps		Spectral Step Offset		No. Spatial Steps		Spatial Step Offset
	1	ALONG SLIT NOD								
Pointing Verification	#	PV Readout Pattern	PV Groups/Int	PV Integrations/Exp	PV Total Integrations	PV Exposures/Dith	PV Total Dithers	PV Total Exposure Time	Optional ETC ID	Filter
	1	FASTR1	34	3	3	1	1	288.604		F560W

Proposal 8051 - Observation 10 - MIRI Spectroscopic survey at $z \sim 10$: Insights into the Nature of Primordial Galaxies

Spectral Elements	#	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Exposures/Dith	Total Dithers	Total Exposure Time	Optional ETC ID
	1	FASTR1	149	8	16	1	2	6654.546	220843
Special Requirements	Background Limited. Background no more than 20th percentile above minimum								
	Group Observations 9, 10, Non-interruptible								

Proposal 8051 - Observation 11 - MIRI Spectroscopic survey at z~10: Insights into the Nature of Primordial Galaxies

Observation	Proposal 8051, Observation 11									Wed May 20 20:00:12 GMT 2026
	Diagnostic Status: Warning									
	Observing Template: MIRI Low Resolution Spectroscopy									
Diagnostics	(Visit 11:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.									
Fixed Targets	#	Name	Target Coordinates		Targ. Coord. Corrections			Miscellaneous		
	(5)	Uncover-26185	RA: 00 14 16.0970 (3.5670708d) Dec: -30 22 40.30 (-30.37786d) Equinox: J2000							
	Comments: Category=Galaxy Description=[Compact galaxies, Emission line galaxies, High-redshift galaxies, Starburst galaxies] Extended=NO									
Acquisition	#	Target	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID	
	1	12 Uncover-26185-TA	F560W	FAST	4	1	1	11.1	221225	
Template	Subarray					Obtain Verification Image?				
	FULL					true				
Mosaic	Rows	Columns	Row Overlap %		Column Overlap %	Row shift (deg)	Column shift (deg)		Tile Order	
	1	2	0.0		80.0	0.0	0.0		DEFAULT	
Dithers	#	Dither Type		No. Spectral Steps		Spectral Step Offset		No. Spatial Steps		Spatial Step Offset
	1	ALONG SLIT NOD								
Pointing Verification	#	PV Readout Pattern	PV Groups/Int	PV Integrations/Exp	PV Total Integrations	PV Exposures/Dith	PV Total Dithers	PV Total Exposure Time	Optional ETC ID	Filter
	1	FASTR1	34	3	3	1	1	288.604		F560W

Proposal 8051 - Observation 11 - MIRI Spectroscopic survey at z~10: Insights into the Nature of Primordial Galaxies

Spectral Elements	#	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Exposures/Dith	Total Dithers	Total Exposure Time	Optional ETC ID
	1	FASTR1	149	10	20	1	2	8319.57	220843
Special Requirements	Aperture PA Range 36.75797 to 64.75797 Degrees (V3 32.0 to 60.0) Background Limited. Background no more than 50th percentile above minimum Group Observations 11, 12, 13, Non-interruptible								

Proposal 8051 - Observation 12 - MIRI Spectroscopic survey at z~10: Insights into the Nature of Primordial Galaxies

Observation	Proposal 8051, Observation 12									Wed May 20 20:00:12 GMT 2026
	Diagnostic Status: Warning									
	Observing Template: MIRI Low Resolution Spectroscopy									
Diagnostics	(Visit 12:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.									
Fixed Targets	#	Name	Target Coordinates		Targ. Coord. Corrections			Miscellaneous		
	(5)	Uncover-26185	RA: 00 14 16.0970 (3.5670708d) Dec: -30 22 40.30 (-30.37786d) Equinox: J2000							
	Comments: Category=Galaxy Description=[Compact galaxies, Emission line galaxies, High-redshift galaxies, Starburst galaxies] Extended=NO									
Acquisition	#	Target	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID	
	1	12 Uncover-26185-TA	F560W	FAST	4	1	1	11.1	221225	
Template	Subarray					Obtain Verification Image?				
	FULL					true				
Mosaic	Rows	Columns	Row Overlap %		Column Overlap %	Row shift (deg)	Column shift (deg)		Tile Order	
	1	2	0.0		80.0	0.0	0.0		DEFAULT	
Dithers	#	Dither Type		No. Spectral Steps		Spectral Step Offset		No. Spatial Steps		Spatial Step Offset
	1	ALONG SLIT NOD								
Pointing Verification	#	PV Readout Pattern	PV Groups/Int	PV Integrations/Exp	PV Total Integrations	PV Exposures/Dith	PV Total Dithers	PV Total Exposure Time	Optional ETC ID	Filter
	1	FASTR1	34	3	3	1	1	288.604		F560W

Proposal 8051 - Observation 12 - MIRI Spectroscopic survey at $z \sim 10$: Insights into the Nature of Primordial Galaxies

Spectral Elements	#	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Exposures/Dith	Total Dithers	Total Exposure Time	Optional ETC ID
	1	FASTR1	149	10	20	1	2	8319.57	220843
Special Requirements	Aperture PA Range 36.75797 to 64.75797 Degrees (V3 32.0 to 60.0) Background Limited. Background no more than 50th percentile above minimum Group Observations 11, 12, 13, Non-interruptible								

Proposal 8051 - Observation 13 - MIRI Spectroscopic survey at z~10: Insights into the Nature of Primordial Galaxies

Observation	Proposal 8051, Observation 13									Wed May 20 20:00:12 GMT 2026		
	Diagnostic Status: Warning											
	Observing Template: MIRI Low Resolution Spectroscopy											
Diagnostics	(Visit 13:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous			
	(5)	Uncover-26185	RA: 00 14 16.0970 (3.5670708d) Dec: -30 22 40.30 (-30.37786d) Equinox: J2000									
	Comments: Category=Galaxy Description=[Compact galaxies, Emission line galaxies, High-redshift galaxies, Starburst galaxies] Extended=NO											
Acquisition	#	Target	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID			
	1	12 Uncover-26185-TA	F560W	FAST	4	1	1	11.1	221225			
Template	Subarray					Obtain Verification Image?						
	FULL					true						
Mosaic	Rows	Columns	Row Overlap %		Column Overlap %	Row shift (deg)		Column shift (deg)	Tile Order			
	1	2	0.0		80.0	0.0		0.0	DEFAULT			
Dithers	#	Dither Type		No. Spectral Steps		Spectral Step Offset		No. Spatial Steps		Spatial Step Offset		
	1	ALONG SLIT NOD										
Pointing Verification	#	PV Readout Pattern	PV Groups/Int	PV Integrations/Exp	PV Total Integrations	PV Exposures/Dith	PV Total Dithers	PV Total Exposure Time	Optional ETC ID	Filter		
	1	FASTR1	34	3	3	1	1	288.604		F560W		

Proposal 8051 - Observation 13 - MIRI Spectroscopic survey at z~10: Insights into the Nature of Primordial Galaxies

Spectral Elements	#	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Exposures/Dith	Total Dithers	Total Exposure Time	Optional ETC ID
	1	FASTR1	149	10	20	1	2	8319.57	220843
Special Requirements	Aperture PA Range 36.75797 to 64.75797 Degrees (V3 32.0 to 60.0) Background Limited. Background no more than 50th percentile above minimum Group Observations 11, 12, 13, Non-interruptible								

Proposal 8051 - Observation 14 - MIRI Spectroscopic survey at z~10: Insights into the Nature of Primordial Galaxies

Observation	Proposal 8051, Observation 14										Wed May 20 20:00:12 GMT 2026	
	Diagnostic Status: Warning											
	Observing Template: MIRI Low Resolution Spectroscopy											
Diagnostics	(Visit 14:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous		
	(6)	CEERS_99715	RA: 14 19 14.8400 (214.8118333d) Dec: +52 44 13.60 (52.73711d) Equinox: J2000									
	Comments: Category=Galaxy Description=[Compact galaxies, Emission line galaxies, High-redshift galaxies, Starburst galaxies] Extended=NO											
Acquisition	#	Target										
	1	NONE										
Template	AcqFilter	Subarray				Obtain Verification Image?						
	F560W	FULL				true						
Mosaic	Rows	Columns	Row Overlap %		Column Overlap %		Row shift (deg)		Column shift (deg)	Tile Order		
	1	2	0.0		80.0		0.0		0.0	DEFAULT		
Dithers	#	Dither Type		No. Spectral Steps		Spectral Step Offset		No. Spatial Steps		Spatial Step Offset		
	1	ALONG SLIT NOD										
Pointing Verification	#	PV Readout Pattern	PV Groups/Int	PV Integrations/Exp	PV Total Integrations	PV Exposures/Dith	PV Total Dithers	PV Total Exposure Time	Optional ETC ID	Filter		
	1	FASTR1	34	3	3	1	1	288.604		F560W		

Proposal 8051 - Observation 14 - MIRI Spectroscopic survey at z~10: Insights into the Nature of Primordial Galaxies

Spectral Elements	#	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Exposures/Dith	Total Dithers	Total Exposure Time	Optional ETC ID
	1	FASTR1	149	14	28	1	2	11649.618	220843
Special Requirements	After Date 20-NOV-2025:00:00:00 Background Limited. Background no more than 20th percentile above minimum								

Proposal 8051 - Observation 15 - MIRI Spectroscopic survey at z~10: Insights into the Nature of Primordial Galaxies

Observation	Proposal 8051, Observation 15									Wed May 20 20:00:12 GMT 2026		
	Diagnostic Status: Warning											
	Observing Template: MIRI Low Resolution Spectroscopy											
Diagnostics	(Visit 15:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous			
	(7)	Uncover-13151	RA: 00 14 22.2003 (3.5925013d) Dec: -30 24 5.27 (-30.40146d) Equinox: J2000									
	Comments: Category=Galaxy Description=[Compact galaxies, Emission line galaxies, High-redshift galaxies, Starburst galaxies] Extended=NO											
Acquisition	#	Target	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID			
	1	13 Uncover-13151-TA	F560W	FAST	10	1	1	27.75	221225			
Template	Subarray					Obtain Verification Image?						
	FULL					true						
Mosaic	Rows	Columns	Row Overlap %		Column Overlap %	Row shift (deg)		Column shift (deg)	Tile Order			
	1	2	0.0		80.0	0.0		0.0	DEFAULT			
Dithers	#	Dither Type		No. Spectral Steps		Spectral Step Offset		No. Spatial Steps		Spatial Step Offset		
	1	ALONG SLIT NOD										
Pointing Verification	#	PV Readout Pattern	PV Groups/Int	PV Integrations/Exp	PV Total Integrations	PV Exposures/Dith	PV Total Dithers	PV Total Exposure Time	Optional ETC ID	Filter		
	1	FASTR1	34	3	3	1	1	288.604		F560W		

Proposal 8051 - Observation 15 - MIRI Spectroscopic survey at $z \sim 10$: Insights into the Nature of Primordial Galaxies

Spectral Elements	#	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Exposures/Dith	Total Dithers	Total Exposure Time	Optional ETC ID
	1	FASTR1	149	11	22	1	2	9152.082	220843
Special Requirements	Background Limited. Background no more than 20th percentile above minimum								
	Group Observations 15, 16, Non-interruptible								

Proposal 8051 - Observation 16 - MIRI Spectroscopic survey at z~10: Insights into the Nature of Primordial Galaxies

Observation	Proposal 8051, Observation 16									Wed May 20 20:00:12 GMT 2026
	Diagnostic Status: Warning									
	Observing Template: MIRI Low Resolution Spectroscopy									
Diagnostics	(Visit 16:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.									
Fixed Targets	#	Name	Target Coordinates		Targ. Coord. Corrections			Miscellaneous		
	(7)	Uncover-13151	RA: 00 14 22.2003 (3.5925013d) Dec: -30 24 5.27 (-30.40146d) Equinox: J2000							
	Comments: Category=Galaxy Description=[Compact galaxies, Emission line galaxies, High-redshift galaxies, Starburst galaxies] Extended=NO									
Acquisition	#	Target	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID	
	1	13 Uncover-13151-TA	F560W	FAST	10	1	1	27.75	221225	
Template	Subarray					Obtain Verification Image?				
	FULL					true				
Mosaic	Rows	Columns	Row Overlap %		Column Overlap %	Row shift (deg)	Column shift (deg)	Tile Order		
	1	2	0.0		80.0	0.0	0.0	DEFAULT		
Dithers	#	Dither Type		No. Spectral Steps		Spectral Step Offset		No. Spatial Steps		Spatial Step Offset
	1	ALONG SLIT NOD								
Pointing Verification	#	PV Readout Pattern	PV Groups/Int	PV Integrations/Exp	PV Total Integrations	PV Exposures/Dith	PV Total Dithers	PV Total Exposure Time	Optional ETC ID	Filter
	1	FASTR1	34	3	3	1	1	288.604		F560W

Proposal 8051 - Observation 16 - MIRI Spectroscopic survey at $z \sim 10$: Insights into the Nature of Primordial Galaxies

Spectral Elements	#	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Exposures/Dith	Total Dithers	Total Exposure Time	Optional ETC ID
	1	FASTR1	149	10	20	1	2	8319.57	220843
Special Requirements	Background Limited. Background no more than 20th percentile above minimum								
	Group Observations 15, 16, Non-interruptible								

Proposal 8051 - Observation 17 - MIRI Spectroscopic survey at z~10: Insights into the Nature of Primordial Galaxies

Observation	Proposal 8051, Observation 17									Wed May 20 20:00:12 GMT 2026
	Diagnostic Status: Warning									
	Observing Template: MIRI Low Resolution Spectroscopy									
Diagnostics	(Visit 17:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.									
Fixed Targets	#	Name	Target Coordinates		Targ. Coord. Corrections			Miscellaneous		
	(8)	GHZ1	RA: 00 14 2.8630 (3.5119292d) Dec: -30 22 18.69 (-30.37186d) Equinox: J2000							
	Comments: Category=Galaxy Description=[Compact galaxies, Emission line galaxies, High-redshift galaxies, Starburst galaxies] Extended=NO									
Acquisition	#	Target	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID	
	1	15 GHZ1-TA	F560W	FAST	4	1	1	11.1	221225	
Template	Subarray					Obtain Verification Image?				
	FULL					true				
Mosaic	Rows	Columns	Row Overlap %		Column Overlap %	Row shift (deg)	Column shift (deg)		Tile Order	
	1	2	0.0		80.0	0.0	0.0		DEFAULT	
Dithers	#	Dither Type		No. Spectral Steps		Spectral Step Offset		No. Spatial Steps		Spatial Step Offset
	1	ALONG SLIT NOD								
Pointing Verification	#	PV Readout Pattern	PV Groups/Int	PV Integrations/Exp	PV Total Integrations	PV Exposures/Dith	PV Total Dithers	PV Total Exposure Time	Optional ETC ID	Filter
	1	FASTR1	34	3	3	1	1	288.604		F560W

Proposal 8051 - Observation 17 - MIRI Spectroscopic survey at z~10: Insights into the Nature of Primordial Galaxies

Spectral Elements	#	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Exposures/Dith	Total Dithers	Total Exposure Time	Optional ETC ID
	1	FASTR1	149	14	28	1	2	11649.618	220843
Special Requirements	After Date 2025.218:00:00:00 Background Limited. Background no more than 20th percentile above minimum								

Proposal 8051 - Observation 18 - MIRI Spectroscopic survey at z~10: Insights into the Nature of Primordial Galaxies

Observation	Proposal 8051, Observation 18 Wed May 20 20:00:12 GMT 2026 Diagnostic Status: Warning Observing Template: MIRI Low Resolution Spectroscopy									
	(Visit 18:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.									
Fixed Targets	#	Name	Target Coordinates		Targ. Coord. Corrections			Miscellaneous		
	(9)	GN-55757-1055757	RA: 12 36 52.2449 (189.2176871d) Dec: +62 11 58.17 (62.19949d) Equinox: J2000							
	<i>Comments:</i> <i>Category=Galaxy</i> <i>Description=[Compact galaxies, Emission line galaxies, High-redshift galaxies, Starburst galaxies]</i> <i>Extended=NO</i>									
Acquisition	#									Target
	1									NONE
Template	AcqFilter	Subarray				Obtain Verification Image?				
	F560W	FULL				true				
Mosaic	Rows	Columns	Row Overlap %	Column Overlap %	Row shift (deg)	Column shift (deg)	Tile Order			
	1	2	0.0	80.0	0.0	0.0	DEFAULT			
Dithers	#	Dither Type	No. Spectral Steps	Spectral Step Offset	No. Spatial Steps	Spatial Step Offset				
	1	ALONG SLIT NOD								
Pointing Verification	#	PV Readout Pattern	PV Groups/Int	PV Integrations/Exp	PV Total Integrations	PV Exposures/Dith	PV Total Dithers	PV Total Exposure Time	Optional ETC ID	Filter
	1	FASTR1	34	3	3	1	1	288.604		F560W

Proposal 8051 - Observation 18 - MIRI Spectroscopic survey at $z \sim 10$: Insights into the Nature of Primordial Galaxies

Spectral Elements	#	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Exposures/Dith	Total Dithers	Total Exposure Time	Optional ETC ID
	1	FASTR1	149	12	24	1	2	9984.594	220843
Special Requirements	After Date 20-NOV-2025:00:00:00 Aperture PA Range 144.75797 to 164.75797 Degrees (V3 140.0 to 160.0) Aperture PA Range 184.75797 to 224.75797 Degrees (V3 180.0 to 220.0) Aperture PA Range 264.75797 to 274.75797 Degrees (V3 260.0 to 270.0) Background Limited. Background no more than 20th percentile above minimum Group Observations 18, 19, Non-interruptible								

Proposal 8051 - Observation 19 - MIRI Spectroscopic survey at z~10: Insights into the Nature of Primordial Galaxies

Observation	Proposal 8051, Observation 19										Wed May 20 20:00:12 GMT 2026
	Diagnostic Status: Warning										
	Observing Template: MIRI Low Resolution Spectroscopy										
Diagnostics	(Visit 19:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous	
	(9)	GN-55757-1055757	RA: 12 36 52.2449 (189.2176871d)								
			Dec: +62 11 58.17 (62.19949d)								
			Equinox: J2000								
		Comments: Category=Galaxy Description=[Compact galaxies, Emission line galaxies, High-redshift galaxies, Starburst galaxies] Extended=NO									
Acquisition	#	Target									
	1	NONE									
Template	AcqFilter	Subarray				Obtain Verification Image?					
	F560W	FULL				true					
Mosaic	Rows	Columns	Row Overlap %		Column Overlap %		Row shift (deg)		Column shift (deg)	Tile Order	
	1	2	0.0		80.0		0.0		0.0	DEFAULT	
Dithers	#	Dither Type		No. Spectral Steps		Spectral Step Offset		No. Spatial Steps		Spatial Step Offset	
	1	ALONG SLIT NOD									
Pointing Verification	#	PV Readout Pattern	PV Groups/Int	PV Integrations/Exp	PV Total Integrations	PV Exposures/Dith	PV Total Dithers	PV Total Exposure Time	Optional ETC ID	Filter	
	1	FASTR1	34	3	3	1	1	288.604		F560W	

Proposal 8051 - Observation 19 - MIRI Spectroscopic survey at $z \sim 10$: Insights into the Nature of Primordial Galaxies

Spectral Elements	#	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Exposures/Dith	Total Dithers	Total Exposure Time	Optional ETC ID
	1	FASTR1	149	12	24	1	2	9984.594	220843
Special Requirements	Aperture PA Range 144.75797 to 164.75797 Degrees (V3 140.0 to 160.0) Aperture PA Range 184.75797 to 224.75797 Degrees (V3 180.0 to 220.0) Aperture PA Range 264.75797 to 274.75797 Degrees (V3 260.0 to 270.0) Background Limited. Background no more than 20th percentile above minimum Group Observations 18, 19, Non-interruptible								

Proposal 8051 - Observation 20 - MIRI Spectroscopic survey at z~10: Insights into the Nature of Primordial Galaxies

Observation	Proposal 8051, Observation 20										Wed May 20 20:00:12 GMT 2026
	Diagnostic Status: Warning										
	Observing Template: MIRI Low Resolution Spectroscopy										
Diagnostics	(Visit 20:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(10)	GS-20088041-207868	RA: 03 32 42.1242 (53.1755175d)								
			Dec: -27 46 50.29 (-27.78064d)								
			Equinox: J2000								
	Comments: Category=Galaxy Description=[Compact galaxies, Emission line galaxies, High-redshift galaxies, Starburst galaxies] Extended=NO										
Acquisition	#										Target
	1										NONE
Template	AcqFilter	Subarray				Obtain Verification Image?					
	F560W	FULL				true					
Mosaic	Rows	Columns	Row Overlap %		Column Overlap %		Row shift (deg)		Column shift (deg)	Tile Order	
	1	2	0.0		80.0		0.0		0.0	DEFAULT	
Dithers	#	Dither Type		No. Spectral Steps		Spectral Step Offset		No. Spatial Steps		Spatial Step Offset	
	1	ALONG SLIT NOD									
Pointing Verification	#	PV Readout Pattern	PV Groups/Int	PV Integrations/Exp	PV Total Integrations	PV Exposures/Dith	PV Total Dithers	PV Total Exposure Time	Optional ETC ID	Filter	
	1	FASTR1	34	3	3	1	1	288.604		F560W	

Proposal 8051 - Observation 20 - MIRI Spectroscopic survey at $z \sim 10$: Insights into the Nature of Primordial Galaxies

Spectral Elements	#	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Exposures/Dith	Total Dithers	Total Exposure Time	Optional ETC ID
	1	FASTR1	149	12	24	1	2	9984.594	220843
Special Requirements	After Date 20-NOV-2025:00:00:00 Aperture PA Range 234.75797 to 324.75797 Degrees (V3 230.0 to 320.0) Aperture PA Range 354.75797 to 79.75797 Degrees (V3 350.0 to 75.0) Background Limited. Background no more than 20th percentile above minimum Group Observations 20, 21, Non-interruptible								

Proposal 8051 - Observation 21 - MIRI Spectroscopic survey at z~10: Insights into the Nature of Primordial Galaxies

Observation	Proposal 8051, Observation 21										Wed May 20 20:00:12 GMT 2026
	Diagnostic Status: Warning										
	Observing Template: MIRI Low Resolution Spectroscopy										
Diagnostics	(Visit 21:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(10)	GS-20088041-207868	RA: 03 32 42.1242 (53.1755175d) Dec: -27 46 50.29 (-27.78064d) Equinox: J2000								
	Comments: Category=Galaxy Description=[Compact galaxies, Emission line galaxies, High-redshift galaxies, Starburst galaxies] Extended=NO										
Acquisition	#	Target									
	1	NONE									
Template	AcqFilter	Subarray				Obtain Verification Image?					
	F560W	FULL				true					
Mosaic	Rows	Columns	Row Overlap %		Column Overlap %		Row shift (deg)		Column shift (deg)	Tile Order	
	1	2	0.0		80.0		0.0		0.0	DEFAULT	
Dithers	#	Dither Type		No. Spectral Steps		Spectral Step Offset		No. Spatial Steps		Spatial Step Offset	
	1	ALONG SLIT NOD									
Pointing Verification	#	PV Readout Pattern	PV Groups/Int	PV Integrations/Exp	PV Total Integrations	PV Exposures/Dith	PV Total Dithers	PV Total Exposure Time	Optional ETC ID	Filter	
	1	FASTR1	34	3	3	1	1	288.604		F560W	

Proposal 8051 - Observation 21 - MIRI Spectroscopic survey at $z \sim 10$: Insights into the Nature of Primordial Galaxies

Spectral Elements	#	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Exposures/Dith	Total Dithers	Total Exposure Time	Optional ETC ID
	1	FASTR1	149	12	24	1	2	9984.594	220843
Special Requirements	Aperture PA Range 234.75797 to 324.75797 Degrees (V3 230.0 to 320.0) Aperture PA Range 354.75797 to 79.75797 Degrees (V3 350.0 to 75.0) Background Limited. Background no more than 20th percentile above minimum Group Observations 20, 21, Non-interruptible								